Compare Leaf Traversal

AlgoExpert · Very Hard · Binary Trees

The problem

- Do two trees have the same sequence of leaves left-to-right?
- Lazy iterators.

Intuition

- Walk both trees' leaves lazily in parallel so we can stop at the first difference.

Approach

- 1. Iterator per tree with a stack
- 2. Advance to the next leaf
- 3. Compare values
- 4. Both must end together

Java solution

```java
public class CompareLeafTraversal {
    public static boolean compareLeafTraversal(BinaryTree a, BinaryTree b) {
        ArrayDeque<BinaryTree> sa = new ArrayDeque<>(), sb = new ArrayDeque<>();
        sa.push(a); sb.push(b);
        while (true) {
            BinaryTree la = nextLeaf(sa), lb = nextLeaf(sb);
            if (la == null || lb == null) return la == lb;
            if (la.value != lb.value) return false;
        }
    }

    private static BinaryTree nextLeaf(ArrayDeque<BinaryTree> st) {
        while (!st.isEmpty()) {
            BinaryTree n = st.pop();
            if (n.left == null && n.right == null) return n;
            if (n.right != null) st.push(n.right);
            if (n.left != null) st.push(n.left);
        }
        return null;
    }
}
```

Complexity

- Time: O(n + m)
- Space: O(h1 + h2)

Recap

- Walk both trees' leaves lazily in parallel so we can stop at the first difference.
- Complexity: O(n + m)